

Yooshin Kim

Industrial Cybersecurity Researcher

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Research Interests

Industrial Cybersecurity (Industrial Control Systems, Critical Infrastructure)

Anomaly Detection (Time-Series Modeling, Anomaly Diagnosis, Contextual Digital Twins)

Trustworthy AI (Explainability, Robustness, Privacy-Preserving Machine Learning)

Education

Integrated M.S.-Ph.D. Program, Electrical Engineering and Computer Science, **Daegu Gyeongbuk Institute of Science and Technology (DGIST)**, Daegu, South Korea

Feb 2023 - present

- Advisor: [Prof. Donghoon Shin](#)

B.S., School of Undergraduate Studies, **DGIST**, Daegu, South Korea

Feb 2018 - Feb 2023

Exchange Student, **University of California, Berkeley**, Berkeley, CA, USA

Jun 2022 - Aug 2022

Publications

* Equal contribution. † Corresponding author.

Heo, W.*, Youn, S.*, **Kim, Y.**, Chae, C., Shin, D.†. "Tracing the Origins: Legacy Codec Identification in Neural Audio Transcoding." *Annual Conference of the International Speech Communication Association (INTERSPEECH)*.

Sep 2026

Kim, Y., Shin, D.†. "Drift-Aware Memory-Augmented Spatio-Temporal Graph Attention for Industrial Anomaly Detection." *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.

Aug 2026

Kim, Y., Kwon, N., Shin, D.†. "KDPrint: Passive Authentication using Keystroke Dynamics-to-Image Encoding via Standardization." *Computers & Security*, 160, 104725.

Jan 2026

Kim, Y.*, Heo, W.*, Woo, E., Shin, D.†. "LCDTwin: Contextual Digital Twin for Actuator-Sensor Correlation in ICS." *5th International Conference on Robotics, Automation and Artificial Intelligence (RAAI)*.

Dec 2025

Jeong, D.*, **Kim, Y.***, Shin, D.†. "Privacy-Preserving Anomaly Detection with Homomorphic Encryption for Industrial Control Systems in Critical Infrastructure." *IEEE Embedded Systems Letters*, 17(4), 276-279.

Aug 2025

Manuscripts Under Review

Kim, Y., Shin, D.†. "MARQ: A Reference-Conditioned Metric for Time-Series Anomaly Score Interfaces."

2026

Woo, E., **Kim, Y.**, Heo, W., Shin, D.†. "System-aware contextual digital twin for ICS anomaly diagnosis."

2026

Chae, C.*, **Kim, Y.***, Shin, D.†. "ELStego: Exact Latent Steganography via Content-Aware Adaptive Margin and Dithered PIT."

2026

Cho, Y., **Kim, Y.**, Shin, D.†. "A Phased Zero Trust Migration Framework for Resilient Space Systems."

2026

Patents

APPARATUS AND METHOD FOR IMAGE-BASED USER AUTHENTICATION USING INPUT BEHAVIOR INFORMATION, (10-2026-0025183, Pending)

Feb 2026

APPARATUS AND METHOD FOR PRIVACY-PRESERVING ANOMALY DETECTION WITHOUT BOOTSTRAPPING UNDER HOMOMORPHIC ENCRYPTION, (10-2026-0018704, Pending)

Jan 2026

FOLDABLE DELIVERY BOX FOR DRONE, (10-2021-0089356, Granted)

Jul 2021

Honors and Awards

Best Presentation Award, RAAI 2025

Dec 2025

Best Paper Award, Korea Computer Congress

Jul 2025

Dean's List, DGIST

Spring/Fall 2021, Fall 2022